

Theory of AutoAugment

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1 Overall Theory

Up until this paper, every data augmentation was done manually. This paper suggests using reinforcement learning to automate this process, while also helping with finding the best augmentation technique which yields the best accuracy.

To do this, the paper uses the concept of policies. Each policies further contains 5 sub-policies. Each sub-policies contain two components, which further contain three variables, which include augmentation technique, probability and magnitude. This specific way of defining policy is done to increase the policy search space and also to introduce richer transformation via combinations.

Now, the paper defines the search space itself. The RL controller will choose one policy from this search space. The search space has every possible combinations of different operations, probability and magnitude. In total, there are fifteen operations, eleven probability and ten magnitudes. In total, there are about 1.49×10^{32} .

Now that the search space is defined, the paper introduces RL Controller, which is used to find policies. Here, the controller network, which is an LSTM, is the agent. The environment is training CNN using policy, while the action is choosing augmentation policy. The reward is the validation accuracy, while the goal is to maximize validation accuracy.

Following this, the paper defines the environment which the agent interacts with. For this, the child network is defined. The purpose of this network is to check whether a policy is good or bad, and it acts as an evaluator. Instead of using the whole training dataset, we use a small subset to reduce computational costs. The paper chooses WideResNet for this purpose.

Once the child network is trained on a certain policy, the paper finds the validation accuracy. This will be used as the reward. Now, the controller increases the probability of generating such similar policies, while simultaneously reducing the probability of generating policies which perform poorly. Finally, this process is repeated thousands of times, specifically fifteen thousand, to find the best possible policies.

2 RL Controller Theory

The paper uses RNN/LSTM as the controller. Since the policies are sequential, the controller needs memory, and hence LSTM is used. Initially, the LSTM uses the start token. This token is converted into an embedding in a hundred-dimensional vector space. This embedding is the first input.

This input is used to predict three heads. First head predicts the operation, second head predicts the probability and third head predicts magnitude. This is fed into the LSTM again with the reward.

For this case, there has to be a balance of exploration and exploitation. This is why it samples from probability distribution, rather than a deterministic policy. If there is too much exploitation, then controller gets stuck in a local maxima. If there is too much exploration, then training becomes random and expensive.

3 Policy Gradient Theory

The paper uses concept of REINFORCE. This implies, increased probability of generating similar policies if the policy performs well and reducing probability of policy which performs poorly. The controller maximizes the expectation value of reward.

Since the sampling is discrete, it is non-differentiable. Therefore, the paper uses the log probability trick for backpropagation.

The rewards are noisy, due to randomness. Without stabilization, the RL updates become extremely noisy. Thus, the paper uses concept of baseline. The baseline is defined as the moving average of rewards. Instead of raw rewards, advantage defined as difference of reward by baseline. This measures the extent of how better the policy is compared to average. This reduces the variance and RL training becomes relatively easy.